Engine Model(s): PT6A

72-50-03

PT INTERSTAGE STATOR - DESCRIPTION AND OPERATION

1. Description and Operation (Ref. 72-50-00, Fig. 1)

The power turbine stator housing consists of a conical casting, machined with open ends to accommodate the power turbine stator. A flange at the front of the housing is bolted together with a containment ring to the rear inner flange of the exhaust duct, while a machined inner diameter on the housing rear extension encasing the interstage sealing ring(s) which fit over a machined extension at the front end of the compressor turbine shroud housing and provides a mechanical joint and seal between the compressor and power turbines.

Eight thermocouple bosses are provided on the rear of the housing for the temperature sensing probes. The threaded bosses are located at the 1, 2, 3, 4, 8, 9, 10 and 11 o'clock positions on the housing. Clips are welded on the outer face of the housing for location and lockwiring of the T5 thermocouple wiring harness . The interstage baffle assembly is between the compressor turbine assembly and the power turbine, and forms part of the power turbine stator assembly. The baffle prevents dissipation of turbine gases and consequent transmission of heat to the turbine disk faces. The center section of the baffle incorporates lipped flanges on the front and rear faces. The flanges fit over corresponding rotor seal flanges on the turbine disks to provide control of cooling air flow through the bore of the compressor turbine and a hole in the center of the baffle. Cooling air is allowed to bleed through the seal and flange interface to cool the front and rear faces of the compressor and power turbine disks respectively; cooling air is then exhausted into the gas stream at the turbine blade roots.

The power turbine stator assembly consists of an inner ring and an outer ring. Nineteen integrally-cast cobalt or nickel alloy vanes between the rings direct the compressor turbine discharge gases onto the power turbine blades at the most efficient angle and velocity. The outer ring is slotted to mate with lugs in the power turbine stator housing. The interstage baffle assembly is secured to a flange in the centerbore of the inner ring by three retaining plates and rivets.

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